

Navigating the Design Implications for Integrating Virtual-Reality Cues Clinically in Drug Psychotherapy

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ABSTRACT

This study investigated the utilization of virtual scenarios to evoke drug cravings and associated physical responses, revealing that self-reported craving scores did not persist beyond the virtual reality (VR) sessions for the 19 participants. Qualitative findings emphasized the importance of considering contextual factors resembling participants' drug use experiences in managing induced cravings. The study underscores the need for careful scenario design based on the patient's recovery stage to effectively manage the intensity of cravings. These insights contribute to the development of VR scenarios for clinical psychotherapy, offering valuable implications for future research and therapeutic applications.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in collaborative and social computing.**

KEYWORDS

Drug addiction, virtual reality, psychotherapy.

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1 INTRODUCTION

Drug psychotherapy, such as cognitive-behavioral therapy (CET), is widely employed to support helping individuals with drug addiction maintain their abstinence from drugs. Cue exposure therapy (CET), a frequently employed technique by therapists, involves immersing patients in cues associated with drug use and has consistently demonstrated its potential across various treatment programs [12, 16]. When integrating CET with psychotherapy, it is crucial that therapists manage the risk of triggering relapse [4, 17] by prioritizing therapeutic considerations over the effectiveness of cue induction. From a patient-centric perspective, the technology solution being designed to support drug psychotherapy must effectively present cues during therapy to help patients recall their previous drug use experiences while minimizing the risk of significantly triggering cravings that could undermine the therapeutic process. Therefore, in the current study, we developed virtual reality (VR) scenarios aimed at inducing cravings by inducing a recollection of past drug use, the results of which provide a focus for collaborative discussions with therapists aimed at enhancing the perception of cravings and associated cues.

However, most of the research in this field did not tailor the design of VR scenarios to expose cues which are carefully designed to meet the therapeutic considerations of drug psychotherapy. Instead, researchers have focused on explicit drug cues that trigger intense

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cravings or the development of algorithms to detect the onset of cravings. In HCI community, there has been very little research on the design of VR scenarios to help patients with drug addiction develop the skills required to detect these cues by themselves. Much of the work on VR-based exposure therapy has focused on its effectiveness in inducing cravings and the development of algorithms to detect craving onset [5, 10, 18, 20]. Recently, Flobak et al. [8] employed user input in the rapid prototyping of 360° VR videos tailored to adolescents seeking to overcome a fear of public speaking. To the best of our knowledge, there has been very little research on the design and use of VR scenarios addressing drug-related cues in support of psychotherapy.

In this work, we conducted an in-lab VR Cue user study aimed at elucidating the effects of cue presentation on the triggering of cravings in virtual scenarios. Our findings revealed that cravings can be triggered by contextual factors that resonate with the drug use experience of participants. Event timing, object placement, and associations among elements were shown to have profound effects on the response of the subject. It is important to match the strength of induced cravings to the recovery stage of the patient, and this is easily achieved within the controlled environment of VR simulations. This study yielded insights for the creation of clinically friendly VR scenarios for use in psychotherapy, with important implications for future research and therapeutic practice.

2 IN-LAB VR CUE STUDY

This in-lab controlled study explored the effects of the effects of cue presentation in virtual scenarios on the triggering of drug cravings and corresponding emotional and physical responses. This study was approved by the Institutional Review Board of Taipei City Hospital (IRB No.: TCHIRB-11009015).

2.1 Participants

A total of 19 participants (18 males) were recruited from Taipei City Hospital, hereafter referred to as "TCH." This research focused on individuals with a history of confronting cues and cravings associated with meth addiction, either through self-management or as part of their treatment regimen. Eligible candidates were identified by therapists at TCH via a meticulous review of the medical records and via snowball sampling of individuals who were currently undergoing outpatient treatment for drug abuse or had completed such treatment.

The candidate pool was subjected to a rigorous screening process conducted by a collaborating psychiatrist to ensure adherence to the following inclusion criteria: 1) DSM-5 criteria for substance use disorders [2], 2) current or past enrollment in outpatient treatment for drug abuse, 3) methamphetamine as the primary drug of abuse, and 4) the ability to read and provide informed consent. The participants ranged in age from 24 to 51 years with a mean age of 36.16 years ($SD = 7.09$ years). All participants had completed education beyond junior high school, most held undergraduate degrees, and some had senior high school or Master's degrees. All participants possessed the fundamental communication skills necessary for meaningful interactions with therapists and psychiatrists. The average self-assessed craving scores were 18.58 ($SD = 19.93$) before the study.

2.2 CravingProbe: The Experimental VR System

This research was based on our CravingProbe VR system, which was designed using Unity with two core modules: (1) craving-inducing and (2) user feedback. This study used a Meta Quest 2 VR headset to immerse participants within a 3D virtual environment. The designs of the three modules are detailed in the following sub-sections.

2.2.1 Craving-inducing module. Drug craving refers to a strong (often irresistible) desire to use drugs, particularly in the absence of such stimuli. Cravings be triggered by external (exteroceptive) cues related to sensory experiences (e.g., sight, sound, smell, and touch), which remind individuals of drug use. Cravings also be triggered by internal (interoceptive) cues (e.g., emotional states and stress) based on individual psychological experiences. One previous study [19] used interviews to identify a range of exteroceptive and interoceptive cues frequently encountered in the daily lives of individuals struggling with meth addiction. We augmented these six scenarios with contextual information displayed via text overlays to guide participants through craving-inducing experiences (Figure 1). To streamline VR scene design, each craving-inducing scenario featured either an exteroceptive or interoceptive cue, which lasted for a uniform duration of 90 s. Exteroceptive cues are featured in three of the scenarios: (1) device, (2) bonus, and (3) hotel room. Interoceptive cues are featured in three of the scenarios: (4) boredom, (5) self-reward, and (6) work pressure. In the following section, we delve into the reasoning behind the inclusion of these scenarios and outline their respective storylines.

- **Device:** Researchers have previously reported that even brief glimpses of drug paraphernalia can induce drug cravings [13]. In the current study, we incorporated a scenario featuring a partially consumed yogurt drink with two straws mimicking the straws used to snort meth.
- **Bonus:** Researchers have previously reported that ready cash can trigger drug binges [6, 7], and our previous research confirmed that some patients experience cravings when they receive their wages. We sought to replicate this real-life scenario by designing an animation in which participants use their fingers to count an unexpected cash bonus of 10,000 NTD (300 USD). The process of counting the money is enhanced by the inclusion of audio feedback of bills being rubbed together.
- **Hotel room:** We sought to replicate environments reminiscent of previous drug use experiences by presenting a hotel room depicting the aftermath of sexual encounter in which meth was used to enhance arousal. The virtual setting includes rumpled towels and discarded food wrappers strewn across the bed and floor as well as hotel pillows and a comforter haphazardly thrown across the bed.
- **Boredom:** Preston et al. [15] established a positive relationship between drug cravings and psychological stress and boredom. In the current study, we sought to emulate this phenomenon by presenting an animation in which the participant views a lock screen notification on their cell phone showing a reminder that it is currently Friday night and a meeting is scheduled for the upcoming Monday. The serenity of the moment is enhanced by the inclusion of audio

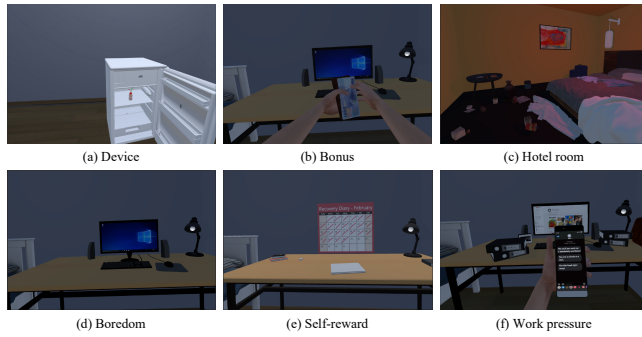


Figure 1: Screenshots of the six VR craving-inducing scenarios.

feedback of background chirping sounds reminiscent of a tranquil summer evening.

- **Self-rewards:** Researchers have reported that addicts often seek to reward themselves for abstinence from their bad habits. In the current study, we sought to recreate the conditions for this behavior by prominently displaying a recovery calendar on the wall in front of the participants as a visual reminder that they had abstained from drug use for 24 days. This message was reinforced by a cellphone animation showing a mobile app indicating the number of days that they have maintained abstinence.
- **Work pressure:** Researchers have demonstrated that strongly negative moods [1] and work-related depression [7, 34, 38] can induce drug addiction. In the current study, we sought to capture this mood by simulating the subject arriving home from work and entering their bedroom late at night (11 PM). A desktop computer displays the YouTube homepage as a subtle indicator of the subject's intention to seek solace in video clips in order to unwind from workday stress. However, another animation shows the subject bringing their phones into view to read an SMS message from their supervisor. In notably impolite language, the supervisor is demanding the the subject redo their work deliverables as a stark reminder of the persistent work pressure (and perhaps on-going failures) associated with their daily routine.

2.2.2 User feedback module. This module was meant to determine the strength of the cravings induced in the subject and realism of a scenario by the simulation. The interface allowed participants to provide responses to survey inquiries using the VR controller, which was configured for ease of input using point-and-click gestures. After experiencing each scenario, the subjects were tasked with assessing their current craving for meth using a Visual Analogue Scale (VAS) [9] and realism of the scenario using a seven-level Likert scale.

2.3 Procedures

This study explored the design of VR scenarios that incorporate carefully curated cues with the aim of giving patients practice in managing craving responses. Note that all of the scenarios were developed based on related literature [19] with clinical applicability

in mind. The study was implemented in three phases: (1) Pre-study, (2) Technology probe, and (3) Post-study interviews.

2.3.1 Pre-study phase (30 minutes). At the outset of the study, the research team provided participants with an overview of the study objectives and introduced the procedures to be followed. Demographic information (age, gender, and educational background) was then obtained after the participants signed informed consent forms.

Study participants were asked to evaluate the strength of their cravings before engaging in the first VR scenario. Baseline assessments pertaining to the strength of cravings before the study involved having patients provide self-assessments of their current meth cravings using a Visual Analogue Scale (VAS) [9]. Following the collection of these data, patients were informed of the study's goals. After this preliminary data was collected, both the patient and their corresponding therapist donned VR headsets. While holding VR controllers in their right hands, the patients were provided instructions on navigating the VR user interface to self-report their responses to the virtual reality scenarios.

2.3.2 Technology probe phase (21 minutes). This phase was executed in five sequential steps: (1) Entry, (2) Tutorial, (3) VR Experience, (4) Recovery, and (5) Exit.

Entry (30 s): The primary objective in this step was to facilitate a seamless transition from the physical to the virtual realm. Upon entering the virtual environment, participants initially found themselves within a virtual replica of the physical room in which the user study was being conducted. This strategic design was meant to minimize disorientation due to perceived spatial disparities between the two environments [11].

Tutorial (30 s): Participants watched a VR tutorial on interactions involving the controller. Their proficiency in operating the devices was then assessed by pressing a button located in the VR environment.

VR experiencing (90 s): Once the participants were comfortable with the VR environment and in a relaxed state, they entered one of the aforementioned scenarios (presented at random). Throughout this immersive experience, participants were free to move their heads and rotate in their chairs, thereby enabling natural and unrestricted exploration of the VR space.

Recovery (90 s): The objective in this step was to allow participants to recovery from the potentially impactful VR scenarios. During this phase, the participants underwent debriefing to assess the intensity of their current cravings that were triggered by the VR scenario that they had just completed. Participants who completed their survey responses before the 90-second mark waited until the full duration had passed.

Exiting (30 s): This stage was the counterpart to the entry stage aimed at ensuring a seamless transition from the virtual world back to reality. Thus, the final VR environment mimicked the room in which they were actually seated and to which they would return [11]. Throughout this procedure, psychotherapists were available to assist participants in removing the VR headset devices.

2.3.3 Post-study interview phase (60 minutes). After the VR session, the post-study drug cravings experienced by the participants were assessed using a VAS [9] and in-depth interviews were conducted one-on-one between patients and research team members to gain

insights into the experiences of the patients during the VR scenarios. Interviewers assessed the degree to which the interview answers aligned with feelings that were self-reported through the user feedback module. Special emphasis was placed on determining the degree to which these recollections of drug use triggered cravings. Patients were asked to give their opinions regarding the design of the VR scenarios and the corresponding impact that this had in conjuring up recollections of previous drug use. They were also asked to provide suggestions for further improvement. The final objective of the interviews was to determine whether insights gained from the VR experience could be used to promote interactions with therapists. Upon the completion of all interview phases, each participant received compensation of approximately NTD \$250 (equivalent to roughly US \$7.8) to cover their travel expenses.

2.4 Measures

Following the completion of each VR scenario, the participants were given a questionnaire aimed at determining the strength of their drug cravings induced by the previous scenario (referred to as the "post-scenario craving score"). Participants were also asked to assess the realism of the scenario (referred to as the "realism score") using a seven-point Likert scale. Craving levels were also assessed before and after the VR experience, which respectively generated *pre-study craving scores* and *post-study craving scores*.

2.5 Data Analysis

Audio recordings from the semi-structured interviews were transcribed to facilitate thematic analysis [3]. We adopted an iterative process of open coding to gain insights into the experiences of the patients during the VR scenarios. Considering the involvement of therapists in interventions aimed at enhancing cue identification and craving perception, we conducted axial coding to establish connections between primary themes with the aim of elucidating how VR scenarios either facilitate or hinder the retrieval of past drug use experiences and how these experiences could potentially enhance patient-therapist interactions. Our analysis revealed issues that should be considered in the design of VR scenarios and application in drug psychotherapy. These insights hold promise for future collaborative efforts between patients and therapists aimed at identifying cues and perceiving drug cravings.

3 RESULTS

3.1 Quantitative Results

The average pre-study and post-study craving scores were 18.58 (SD = 19.93) and 20.16 (SD = 19.28), respectively. These results indicate mild cravings before the study and no statistically significant increase in craving after the study.

After watching individual scenarios, the mean (standard deviation) post-scenario scores were as follows: Device ($m = 18.32$; 18.09), bonus ($m = 16.21$; $std = 24.40$), hotel room ($m = 25.58$; $std = 28.20$), boredom ($m = 32.11$; $std = 30.99$), self-reward ($m = 19.42$; $std = 28.06$), and work pressure ($m = 13.16$; $std = 18.40$). Clearly, the boredom scenario induced the strongest cravings, while work pressure had the least effect on cravings.

The mean and standard deviation scores for realism were as follows: device ($m = 3.61$; $std = 1.76$), bonus ($m = 4.74$; $std = 1.55$),

hotel room ($m = 5.05$; $std = 1.39$), boredom ($m = 4.58$; $std = 1.82$), self-reward ($m = 4.84$; $std = 1.56$), and work pressure ($m = 4.26$; $std = 1.55$). The scenarios can be ranked scenarios in terms of average realism from low to high as follows: Device < work pressure < boredom < bonus < self-reward < hotel room.

3.2 Qualitative Findings

3.2.1 The Importance of triggering elements in creating scenarios that resonate with drug users. Every element in a simulation should prompt the viewer to ponder their own pertinent life experiences. The boredom scenario contained a large number of items commonly associated with weekend boredom, which may explain why this scenario resulted in the highest average post-scenario craving score. For instance, the seemingly innocuous act of checking the time on the phone closely mirrored the experiences of those who had previously prepared and used meth, triggering vivid recollections of the entire process. P08 commented "*[Regarding the boredom scene, how would you describe your overall experience?] The use of my mobile phone in this segment is reminiscent of my past drug use... Just look at the phone. Take a look and throw it aside... Then, I take out my meth pipe and a water bong out and place them nicely in front of me. Eventually, I envision myself using them all through the night... [Why did you assign a craving score of 100 points for this scenario?] I could easily relate to this scenario, in which I would undoubtedly follow my whims.*"

Participants also pointed out that the quantity of some visual elements (e.g., reward in the bonus scenario or the duration of abstinence in the self-reward scenario) did not match their actual experiences. For example, some participants noted that the size of the bonus (in the bonus scenario) was inadequate, considering that they needed to allocate a portion of the money to cover daily expenses, which would leave only enough money for a small amount of meth. They reported that this dampened the intensity of their cravings. P15 commented: "*I believe that the money-counting [self-reward] scenario would have been more effective if the bonus were bigger. Most of the 10,000 NTD would go to living expenses, leaving only 3,000, which is enough for only one gram of meth. This increased my craving to some extent; however, a larger sum of money would have allowed me to buy a lot more, which would have intensified the craving.*"

Some participants noted disparities between the information on the visual elements and their own past experiences. They noted that the information was in line with daily life, but differed from their own experiences. They also claimed that this impeded their ability to empathize with the narrative, which reduced the strength of the craving. P10 commented: "*I found it highly realistic because I think it would resonate with individuals who have experienced similar workplace pressure. However, the way my supervisor communicates with me and pushes me to complete tasks is different, especially during mid-term report submissions when I'm responsible for data verification. Nevertheless, he does expect flawless data presentation and frequently requests updates, which involves continuous inquiries about the situation. This scenario reminded me of when I received similar text messages. I did not have exactly that experience, but I am sure that if had, the craving would have been even stronger.*"

3.2.2 Influences of object placement, timing of events, and element associations on the triggering of cravings. Context refers to the details that encompassing a specific concept, situation, or event. In these scenarios, context refers to object placement, event timing, or associations among elements that clarify the narrative and help to embody the viewer within the virtual environment. Several participants remarked that the erroneous arrangement of objects within the scenarios could hinder the evocation of cravings. For instance, it was pointed out that yogurt drink in the refrigerator was unusual. The arrangement reminded them of drug paraphernalia; however, the fact that it appeared in the refrigerator was not in line with the way they used to deal with such items. Apparently, this counter-intuitive placement of cues hindered the elicitation of cravings. P09 commented: *“[How did you feel when you saw the yogurt drink in the refrigerator?] I wouldn’t store my water bong in the refrigerator. That stuff isn’t normally kept in there - it would be concealed in a corner of a cabinet instead. Putting it the fridge does not align with the habits of regular users, so it looks out of place. You could consider placing it in a corner or on the edge of a bed, to evoke such a sensation.”*

The specific timing of the events depicted in some scenarios (Work scenario) was also shown to affect the triggering of cravings. Considering the time required to come down from a meth high, some participants felt that the timing wasn’t properly aligned. They reported that would have to wait until the effects of the drug had worn off before going to work the next day. P11 commented: *“[Why did you give the Work scenario with a low craving score of 5, when the supervisor sent a stern message? Can you explain?] The key factor here is the timing of the boss’s message. At 11 o’clock in the evening, I wouldn’t even consider using Meth. I might indulge after work (at around 8 or 9 o’clock), or perhaps even after dinner. Then I stay up until 1 or 2 in the morning. A message arriving from the boss at 9:00 might tempt me to take some meth, but at 11 o’clock, no way.”*

The association between objects can make a scenario even more suggestive of drug use. For example, participants use a lighter to heat the crystal meth and make it vaporize. Some participants explicitly mentioned the link between drug paraphernalia and disposable lighters. P18 commented: *“I would recommend placing the yogurt drink directly on the table alongside a lighter. That would be really suggestive. If I saw that, my body would respond with genuine cravings. This would provide a more authentic representation, mimicking a typical scenario.”*

3.2.3 Maintaining induced cravings at a manageable level. Some of the participants favored cues that were subtle or indirect to lessen their lasting impact. They argued that crafting cues to be overly realistic or detailed would drive cravings to unmanageable levels. P04 commented: *“[Your craving score after the Yoghurt scenario was 65. Could you explain why?] Well, the yogurt drink with two straws looked like a water bong for meth. A more vivid representation of drug paraphernalia would have been a really intense trigger, resulting in a really strong craving. Then, the cue would be more than a hint; it would become a direct stimulus and trigger a strong craving.”*

Other participants favored direct cues with greater detail. They viewed the cues in the scenarios from a more practical perspective, believing that a merely glimpse of the yoghurt container with straws would be insufficient to trigger a craving. They appeared more pragmatic in the way that they considered how they would use

the tools. P02 commented: *“From a practical standpoint, I personally believe that just seeing something that looks like a bong would not create a hazardous environment. Using meth requires a specific setup with components, like a glass ball connected to the water bong to hold the meth while it is heated. That yogurt container with straws lacks these essential components - it’s incomplete. I couldn’t just pick it up and use it. I really don’t believe it would trigger an impulse or lead me back to drug use. As I see it, a truly dangerous situation would be the elements almost fully assembled and ready for use. That would pose a serious risk.”*

Other participants had serious misgivings about the inclusion of cues that are directly related to drug use (e.g., syringes and pipes), despite the fact that they would enhanced the realism. Note that the attitude toward the adoption of explicit cues depended largely on the patient’s stage of recovery. Patients with less severe drug cravings felt sure that they could handle these cues. Patients with more severe cravings and a more disorganized daily life would probably be unable to handle the cravings induced by direct cues. Scenarios that lack realism would probably not be taken seriously by individuals who approach things logically. P11 commented: *“If your aim was to enhance the realism of the scenarios, then you should include BD [the brand] syringes somewhere in the scenario, although that could trigger intense cravings. The realism of the cues should be determined by the current state of the patient. The same cues could elicit very different responses in different people. The fidelity of cues should hinges on the individual’s ability to manage the resulting cravings.”*

4 DISCUSSION: THE POTENTIAL OF CRAFTING VR SCENARIOS FOR PSYCHOTHERAPY

In this in-lab study, some patients found it difficult to extrapolate past drug use experiences from the contextual elements presented in the scenarios. As discussed in Section 3.2.3, this largely explains the low realism scores for the device and work pressure scenarios. Essentially, some of the elements failed to resonate with the life experiences of the participants (Section 3.2.1 or the contextual cues were poorly designed (Section 3.2.2).

Considering the ease with which graphic elements can be configured, it should be possible to design an interface that allows the dynamic adjustment of scenario parameters prior to starting the VR scenario experience. For example, the system could provide a set of candidate items from which to choose (e.g., phone messages or devices), based on the strength of responses to stimuli in previous sessions. The therapist should have the ability to personalize scenario parameters; however, it is important not to overwhelm the system with too many settings. In the future, research should be conducted with the aim of identifying the essential settings and a suitable interface by which to tailor scenarios to the patient prior to cue exposure. This is akin to prescribing the right dosage of medicine according to the individual needs.

It is important to consider that the thoughts and cravings triggered by the simulation scenarios can serve as a common foundation for interactions between the patient and therapist. Patient P08 noted that the simulations provided common ground to overcome the difficulties in knowing what to say in previous sessions with a

therapist. Nonetheless, several aspects remain to be explored. Practical therapeutic application of these methods for the treatment of substance use disorder will require further research to identify the scenarios that should or should not be included in each session as well as the length and duration of the simulations. The sessions in the current study were lasted 30-60 minutes [14], in accordance with the average attention span of patients undergoing treatment. It will also be necessary to assess the condition of the patient (perhaps by referencing previous sessions) in order to match the strength of the cues and the level of simulation fidelity to the unique requirements of the patient.

5 CONCLUSION

This study examined how to design virtual scenarios to properly elicit drug cravings in a clinical context. Qualitative analysis showed contextual factors resonating with participants' drug use experiences triggered cravings. Timing, object placement, and element associations influenced craving strength. Controlling craving intensity necessitates careful attention to scenario completeness and nuances, aligning with the patient's recovery stage. This study yielded insights for the development of clinically friendly VR scenarios for use in psychotherapy, with important implications for future research and therapeutic practice.

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